

Montepio Developer Hub

Backstage Guidelines

The purpose of this basic guide is to clarify how to use the Backstage Platform when requesting Git repository, YAML Pipelines, Security Roles, and Library – Variable Groups (OPS and APP) creation for new or migrated applications.

Backstage Template

Go to the [Backstage](#) platform, then on the left menu, select “Create...” to choose the template for new component creation.

For example, “Azure DevOps On Prem APPS workload Repository”:

<https://backstage.platform.montepio.com/create/templates/default/devops-create-onprem-apps-repo>

This Template is intended to provide self-service instantiation of Azure Devops workspaces (repos/pipelines/var-groups/etc) for On Prem APPS.

Tech Stack Page

Devops Project*: Select “Development” option.

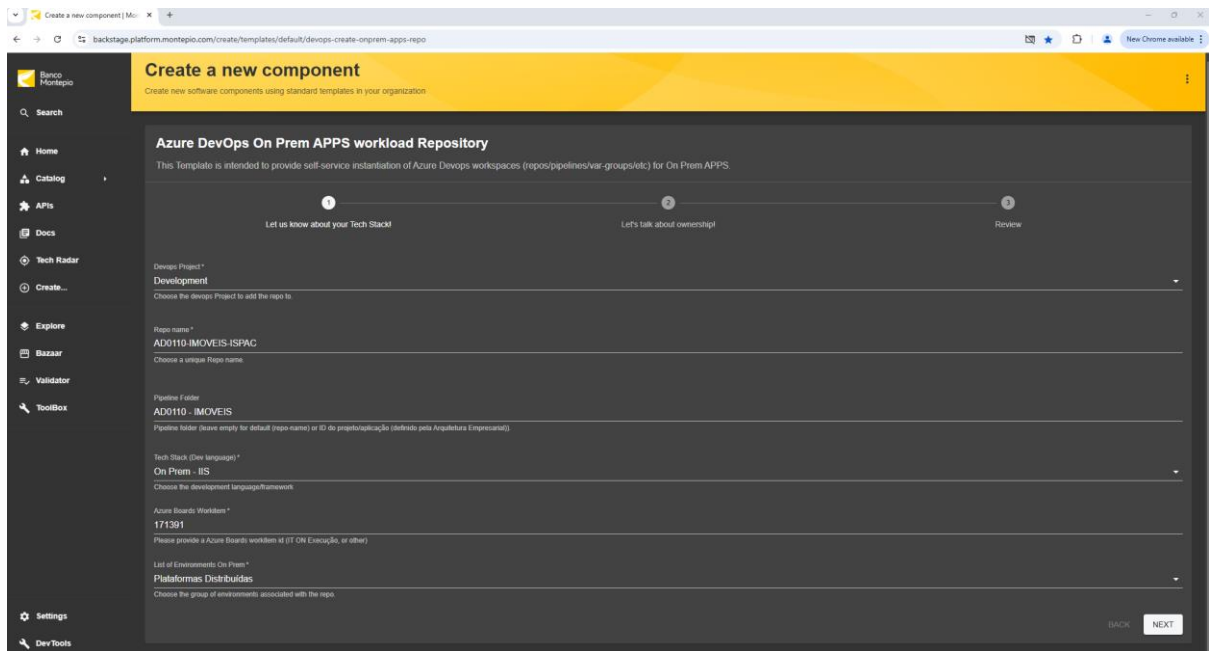
Repo name*: “AD0101-SIRA-WEB” or “AD0101-SIRA-WS” or “AD0101-SIRA-SQL” or “AD0101-SIRA-ISPAC”.

Pipeline Folder: “AD0101 - SIRA”.

Tech Stack (Dev language)*: “On Prem – IIS” or “On Prem – IIS” or “On Prem – DACPAC” or “On Prem – ISPAC”.

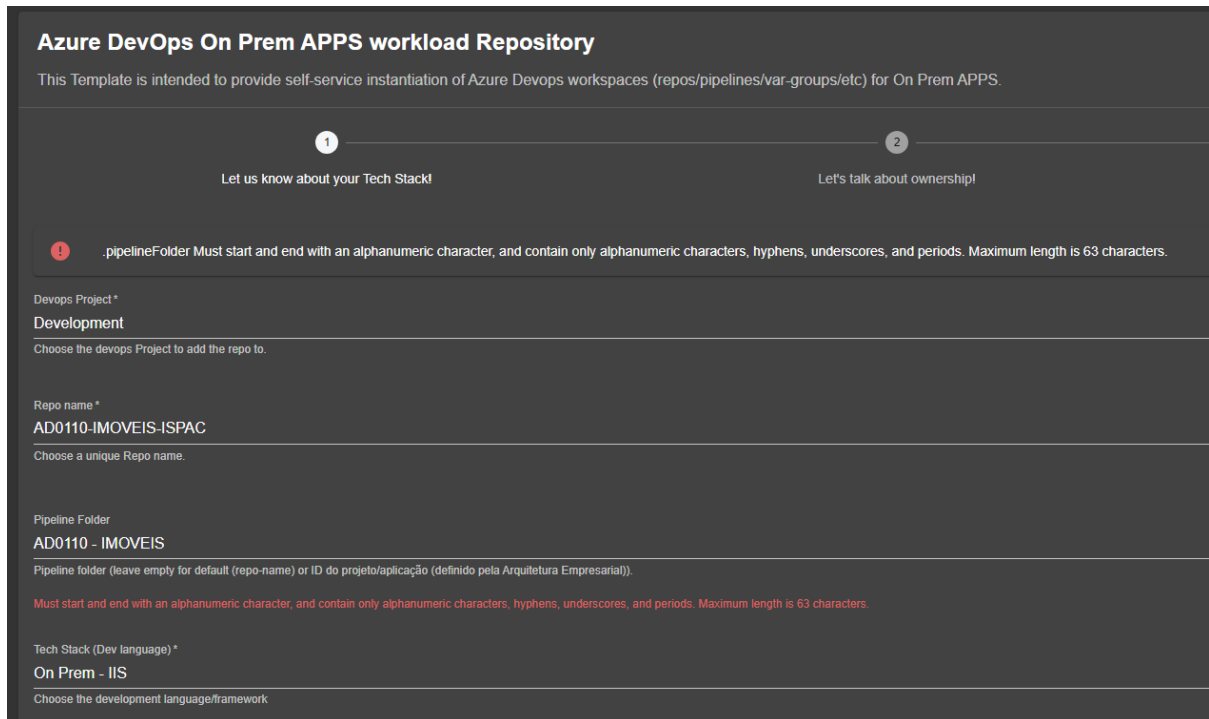
Azure Boards Workitem*: “171391”, *Work Item ID of the task created for DEOS/Plataformas that justifies the Repos creation. You can check this “171391”.*

List of Environments On Prem*: Select “Plataformas Distribuídas” option.



<NEXT>

NOTE: For now, when moving to the next page, we are facing this error:



We must remove the spaces of the “Pipeline Folder” to move on, but this will cause issues on ISPAC repos, OPS variables and environment, then we have to request DOST to update the OPS variables before the first ISPAC deployment.

This was already informed to DEOS/Plataformas and DEOS/Sistemas, and we are still waiting for the normalization of the logic behind this template.

Ownership Page

Dev Team (prefix): “ad0101”.

Área Funcional (repo folder)*: Select “Verticais” option.

Área Funcional Aprovação*: Select “Verticais” option.

The screenshot shows the 'Create a new component' page in the Backstage platform. The page title is 'Create a new component' and the subtitle is 'Create new software components using standard templates in your organization'. The main heading is 'Azure DevOps On Prem APPS workload Repository'. Below this, there is a progress bar with three steps: 'Let us know about your Tech Stack!', 'Let's talk about ownership!', and 'Review'. The 'Review' step is currently active. The form fields are as follows:

- Dev Team (prefix):** (Must match existing DevOps teams -> dev/teams -> Contributors + -> dev/teams -> PR/Approvals) (Only allowed lowercase, numbers, hyphens and underscores) "ad0101"
- Choose the dev team:** (Dropdown menu)
- Area Funcional (repo folder)*:** (Dropdown menu) "Verticais"
- Choose the functional area folder:** (Dropdown menu)
- Area Funcional Aprovação*:** (Dropdown menu) "Verticais"
- Choose the repo functional area:** (Dropdown menu)

At the bottom right, there are 'BACK' and 'REVIEW' buttons.

<REVIEW>

Create a new component
Create new software components using standard templates in your organization

Azure DevOps On Prem APPS workload Repository

This Template is intended to provide self-service instantiation of Azure DevOps workspaces (repos/pipelines/var-groups/etc) for On Prem APPS.

Let us know about your Tech Stack! Let's talk about ownership! **Review**

Repo Name	AD0110-IMOVEIS-ISPAC
Project Name	Development
Pipeline Folder	AD0110-IMOVEIS
Module Source	On Prem - IIS
Workitem	171391
List Environment	Plataformas Distribuidas
Dev Team	ad0101
Functional Area Folder	Verticals
Functional Area Approval	Verticals

BACK **CREATE**

<CREATE>

This request is now sent to DEOS/Plataformas (L.Coelho and A.Padrão) for approval.

Do not forget to message them and share the Work Item created for this request.

Once approved, all the following will be automatically created:

Git repository,

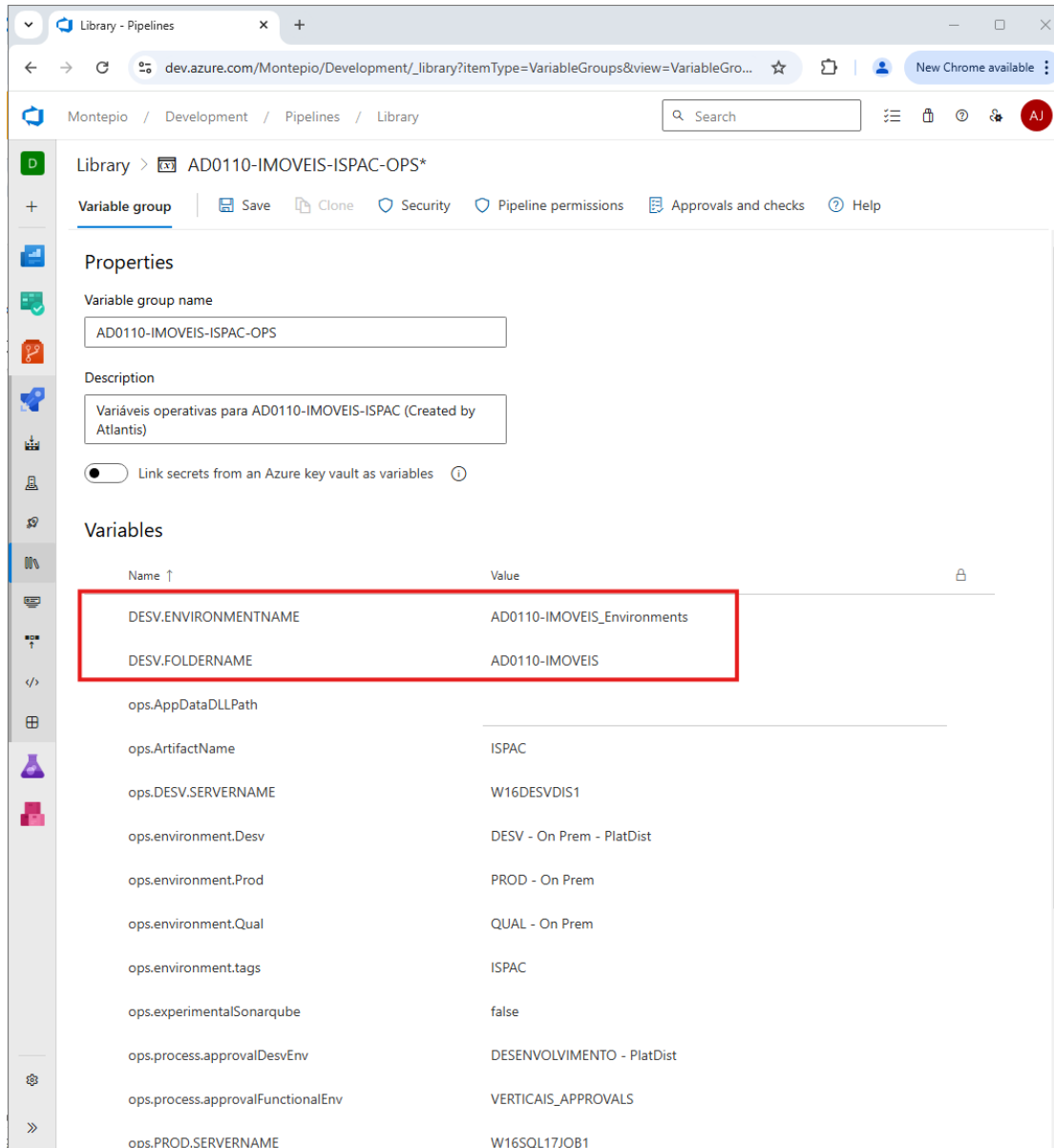
YAML Pipelines,

Security Roles,

and Library – Variable Groups (OPS and APP).

ISPAC Issue

For the ISPAC Template issue, this is what we get created:



The screenshot shows the Azure DevOps interface for a Variable Group named 'AD0110-IMOVEIS-ISPAC-OPS*'. The 'Properties' section displays the variable group name and a description: 'Variáveis operativas para AD0110-IMOVEIS-ISPAC (Created by Atlantis)'. Below this, the 'Variables' section contains a table of variables. Two variables are highlighted with a red box:

Name	Value
DESV.ENVIRONMENTNAME	AD0110-IMOVEIS_Environments
DESV.FOLDERNAME	AD0110-IMOVEIS

Other variables listed include:

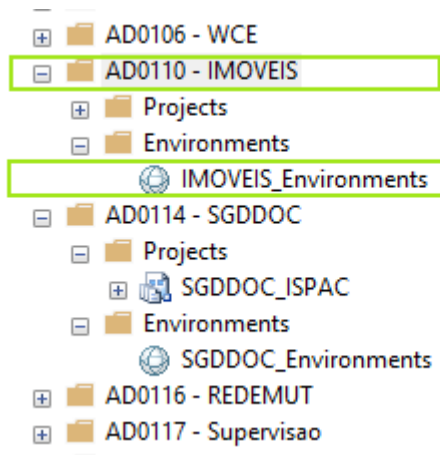
- ops.AppDataDLLPath
- ops.ArtifactName: ISPAC
- ops.DESV.SERVERNAME: W16DESVDIS1
- ops.environment.Desv: DESV - On Prem - PlatDist
- ops.environment.Prod: PROD - On Prem
- ops.environment.Qual: QUAL - On Prem
- ops.environment.tags: ISPAC
- ops.experimentalSonarqube: false
- ops.process.approvalDesvEnv: DESENVOLVIMENTO - PlatDist
- ops.process.approvalFunctionalEnv: VERTICAIS_APPROVALS
- ops.PROD.SERVERNAME: W16SQL17JOB1

And when deploying the application with these values, the Application Folder and Environment are not normalized as they should...



In this case, our team needs to update this manually for normalization.

The Environment name is updated from “*AD0110-IMOVEIS_Environments*” to “*IMOVEIS_Environments*”. And the Application Folder, is now written with spaces near the hyphen as “*AD0110 - IMOVEIS*”:



And we need to request DEOS/Sistemas (L.Palma) to update the related variables on OPS Variable Group, in our example, “*AD0110-IMOVEIS-ISPAC-OPS*”.

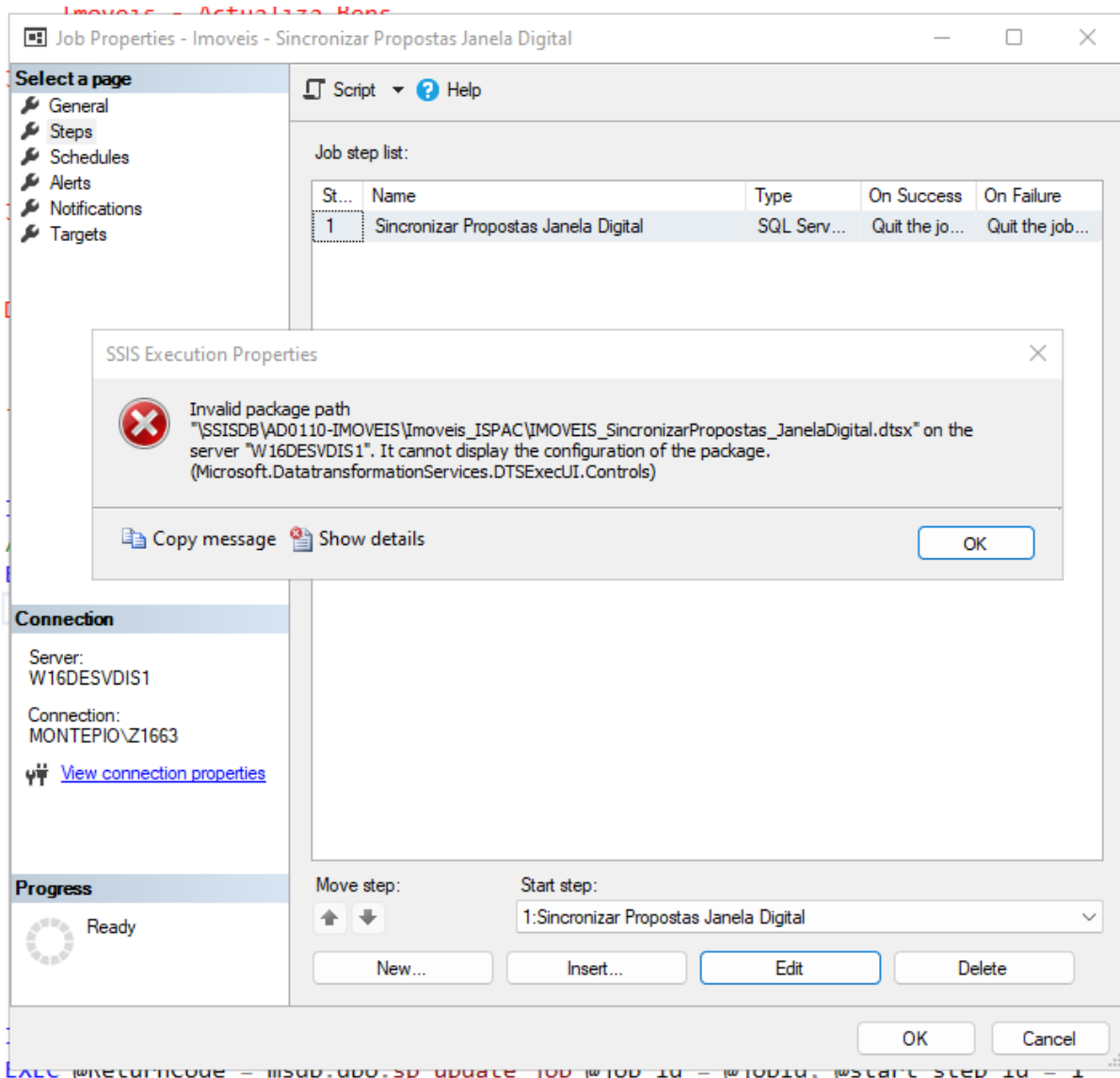
IMPORTANT

If we have jobs already created, we must update all of them, because the jobs have the wrong reference to the DTSXs and Environment:

TITLE: SSIS Execution Properties

*Invalid package path "\SSISDB\AD0110-
IMOVEIS\Imoveis_ISPAC\IMOVEIS_SincronizarPropostas_JanelaDigital.dtsx" on the
server "W16DESVDIS1". It cannot display the configuration of the package.
(Microsoft.DataTransformationServices.DTSExecUI.Controls)*

When opening the Job Step.



Job Step Properties - Sincronizar Propostas Janela Digital

Select a page

- General
- Advanced

Script ? Help

Step name: Sincronizar Propostas Janela Digital

Type: SQL Server Integration Services Package

Run as: ImoveisJobUserProxy

Package Configuration

Package source: SSIS Catalog

Server: W16DESVDIS1

Log on to the server

☒ Use Windows Authentication

☐ Use SQL Server Authentication

User name:

Password:

Package: \SSISDB\AD0110-IMOVEIS\Imoveis_ISPAC\IMOVEIS_SincronizarPropostas_...

Previous Next

OK Cancel

Connection

Server: W16DESVDIS1

Connection: MONTEPIO\Z1663

[View connection properties](#)

Progress

Ready

From:

*\SSISDB\AD0110-
IMOVEIS\Imoveis_ISPAC\IMOVEIS_SincronizarPropostas_JanelaDigital.dtsx*

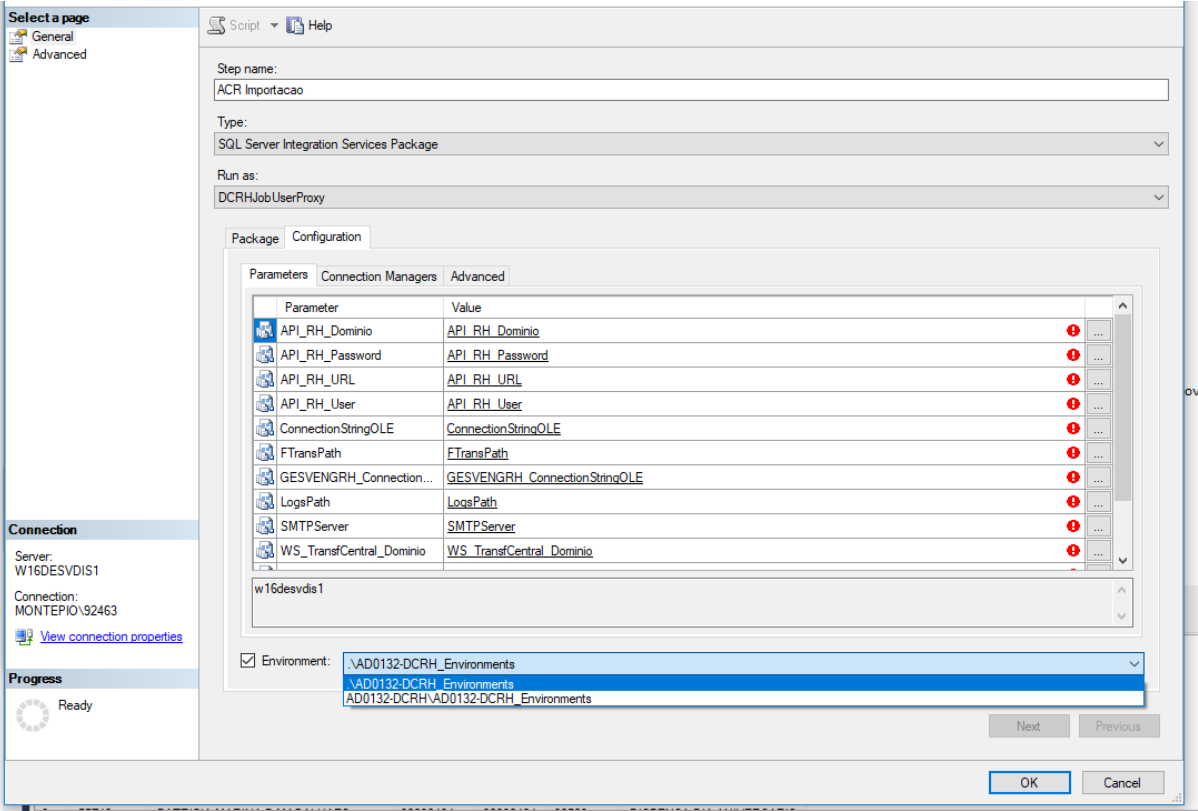
To:

*\SSISDB\AD0110 -
IMOVEIS\Imoveis_ISPAC\IMOVEIS_SincronizarPropostas_JanelaDigital.dtsx*

But the Environment is still bad, because the *[SSISDB].[internal].[environment_references]* table has not been updated with the renaming.

	reference_id	project_id	reference_type	environment_folder_name	environment_name	validation_status	last_validation_time
1	130146	140087	A	AD0110-IMOVEIS	AD0110-IMOVEIS_Environments	N	NULL
2	130147	140087	R	NULL	AD0110-IMOVEIS_Environments	N	NULL

Causing this issue (this is a sample from DCRH after the renaming):



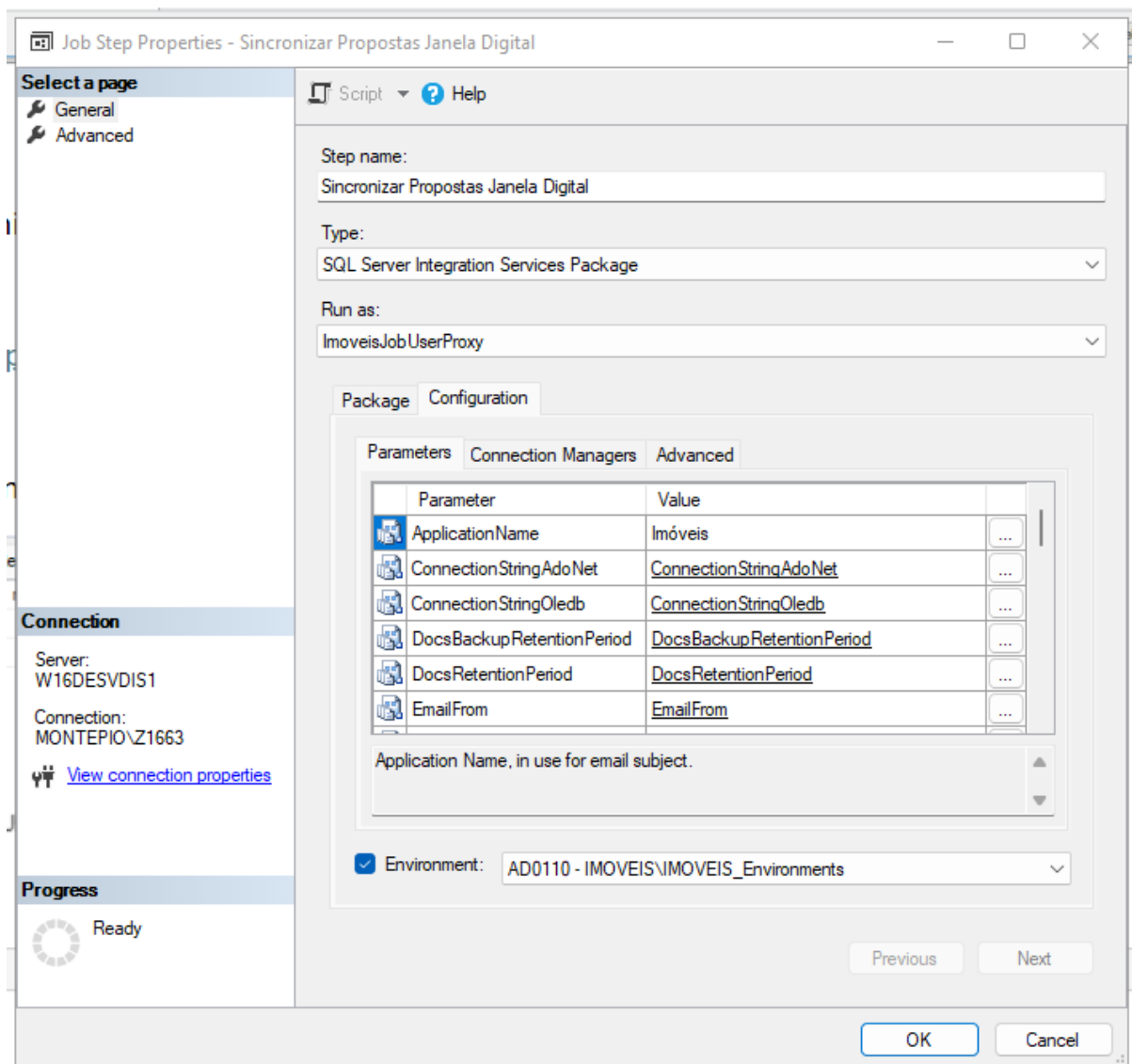
In this case, we should run this script:

Script for Updates on SSISDB Environments Table

Then:

	reference_id	project_id	reference_type	environment_folder_name	environment_name	validation_status	last_validation_time
1	130146	140087	A	AD0110 - IMOVEIS	IMOVEIS_Environments	N	NULL
2	130147	140087	R	NULL	IMOVEIS_Environments	N	NULL

And we are good to update the Job Step via SSMS User Interface:



All this process is now complete, but we can avoid it, following the recommended action below.

RECOMMENDED ACTION

However, when requesting DEOS/Sistemas to act on this before the deployment and before any job creation, we have the variables of “*AD0110-IMOVEIS-ISPAC-OPS*” already normalized, then no updates on our project in SSISDB is necessary:

Library - Pipelines

dev.azure.com/Montepio/Development/_library?itemType=VariableGroups&view=VariableGro...

Montepio / Development / Pipelines / Library

Library > AD0110-IMOVEIS-ISPAC-OPS

Variable group | Save | Clone | Security | Pipeline permissions | Approvals and checks | Help

Properties

Variable group name

AD0110-IMOVEIS-ISPAC-OPS

Description

Variáveis operativas para AD0110-IMOVEIS-ISPAC (Created by Atlantis)

☐ Link secrets from an Azure key vault as variables ⓘ

Variables

Name ↑	Value	🔒
DESV.ENVIRONMENTNAME	IMOVEIS_Environments	
DESV.FOLDERNAME	AD0110 - IMOVEIS	
ops.AppDataDLLPath		
ops.ArtifactName	ISPAC	
ops.DESV.SERVERNAME	W16DESVDIS1	
ops.environment.Desv	DESV - On Prem - PlatDist	
ops.environment.Prod	PROD - On Prem	
ops.environment.Qual	QUAL - On Prem	
ops.environment.tags	ISPAC	
ops.experimentalSonarqube	false	
ops.process.approvalDesvEnv	DESENVOLVIMENTO - PlatDist	
ops.process.approvalFunctionalEnv	VERTICAIS_APPROVALS	
ops.PROD.SERVERNAME	W16SQL17JOB1	

For the IMOVEIS – ISPAC on this practical example, the ENVIRONMENTNAME and FOLDERNAME are now correct, and the deployment will succeed at first attempt, avoiding any further maintenance from our end.